

Manpreet Singh

manpreet.s.cse@gmail.com / Halifax, NS

Education

Master of Computer Science

Jan 2023 – Dec 2024

Dalhousie University – *Halifax, NS, Canada*

Bachelor of Computer Applications

May 2016 – May 2019

Guru Nanak Dev University College – *Jalandhar, India*

Research Experience

CS Graduate Student

Jan 2023 – Dec 2024

Dalhousie University – *Halifax, NS, Canada*

- Explored language model interpretability by applying Integrated Gradients to identify salient neurons, generating model-based explanations of learned textual patterns by neurons, and evaluating these explanations to assess the self-explanatory capacity of language models.
 - Evaluated manifold learning techniques for dimensionality reduction of NLP token embeddings, visualizing results to identify model-learned biases and assess algorithm performance in capturing linguistic nuances.
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Work Experience

AI Solutions Engineer

Jul 2019 – Oct 2022

Tata Consultancy Services – *Chennai, India*

- Built an **NLP chatbot using Rasa** to streamline coding tasks, resulting in a 25% increase in developer productivity.
- Engineered a Computer Vision application that transforms **2D architectural blueprints into 3D** visual models, revolutionizing how architects and clients interact with plans.
- Developed a GPT-3 powered bot designed to translate **English text into SQL queries**, featuring dynamic prompt engineering for optimized database interactions. The bot was specially tailored to interface with a **custom database schema**, providing a highly user-friendly and context-aware querying experience.

- Conducted lectures on Python and Artificial Intelligence, **mentored new joiners**, and **supervised project** timelines and deliverables as part of the onboarding process for over **100** junior engineers.
- Excelled in an initial 20-week intensive training program focused on AI and machine learning, securing a rank within the **top 0.5%** among a cohort of 1000 trainees. This strong foundation contributed to my subsequent contributions in machine learning and AI projects, including chatbot and Computer Vision application development.

Recognition and Awards

- Showcased a cutting-edge Neural Style Transfer project during the visit of India's Honorable Education Minister to Ignite, TCS, receiving high-level recognition for innovation in AI.
- Secured first place in the Plasma Quiz L.K. college, Jalandhar, showcasing extensive knowledge in science and technology.

Skills

Programming Languages

Python	JavaScript	Java	C	SQL
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Machine Learning & AI

Supervised Learning	Unsupervised Learning	Deep Learning	Neural Network Interpretation	NLP	NLU	
Explainable AI	Transformers	LLMs	Chatbot Development	Prompt Engineering	Computer Vision	GANs
CNN	PyTorch	TensorFlow	Keras	Scikit-learn	Rasa	

Data Science

Data Analysis	Data Visualization	Statistics	Pandas	Numpy	Matplotlib	Seaborn	EDA
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Software Development & DevOps

Git	Docker	RESTful APIs	Flask	Django
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Management & Leadership

Agile	Project Management	Mentoring	Team Leadership	Onboarding
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